



08/17/2025

Dr. Jen Zen Ho
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Dear Graduate Admissions Committee,

It is with great pleasure to acknowledge the valuable contribution of Mr. Swapnil Kumar, who worked as a Machine Learning Specialist for Stanford University Flame AI Challenge 2024. The entire programme was scheduled from 23 September to 18 October 2024. The mission of the programme is to foster a dynamic forum for exchanging ideas, cutting-edge data, advanced methods, and models related to ML techniques for fluid dynamics, turbulence, and environmental science, applications crucial to the development of sustainable energy solutions, climate resilience, wildfire mitigation, and safety systems.

Swapnil worked with Professor Matthias Ihme and Stanford FX Lab team members. Swapnil gave the lecture titled Convolutional LSTM and PredRNN Models to the participants and taught multiple participants during the office hours. Swapnil discussed Convolutional LSTM model, PredRNN model, Physics Informed Neural Network model and Surrogate Model (Multi-Fidelity Deep Neural Network) models with the participants during the office hours. He has also helped the participants in debugging and simplifying the code and training the Machine learning model on Kaggle during the office hours. During the office hours, he did teach an average of 7 participants in every session and answered an average of 20 questions in every session.

Swapnil has been an outstanding contributor to the project of significant scientific and societal importance, focused on applying advanced machine learning techniques to predict and mitigate the devastating impacts of wildfires. In this role, he leveraged expertise in AI/ML to address a complex, high-stakes environmental challenge, producing results with direct implications for both research and real-world application. A key aspect of Swapnil contribution was the application of state-of-the-art sequence prediction methods to complex, high-dimensional datasets, enabling more accurate forecasting of wildfire spread and intensity. This work not only demanded technical mastery of advanced machine learning architectures, but also an ability to adapt and innovate in the face of challenging and dynamic data conditions. Beyond the technical

execution, Swapnil actively engaged with a global network of leading scientists and machine learning experts. Through this collaboration, Swapnil also exchanged ideas, refined methodologies, and contributed to a vibrant community dedicated to harnessing AI for positive environmental impact.

We thank Swapnil for his outstanding contributions and for helping to advance the Stanford mission of advancing Machine Learning research in promoting cross-cutting scientific exchange and innovation.

Sincerely,

A handwritten signature in black ink, appearing to read "Ho Jen Zen". The signature is stylized with a large "H" and a cursive "Jen Zen".

Dr. Jen Zen Ho
Postdoctoral Scholar